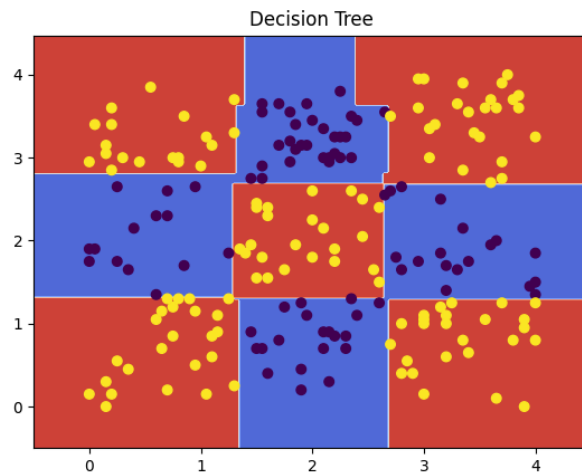
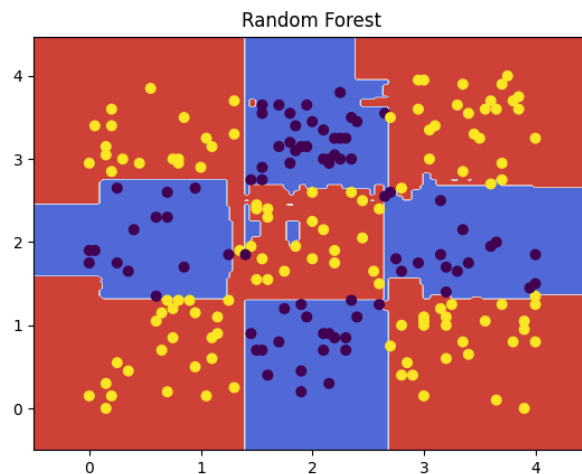
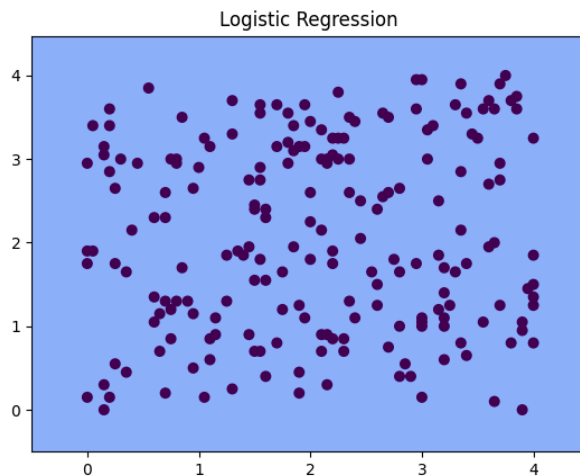




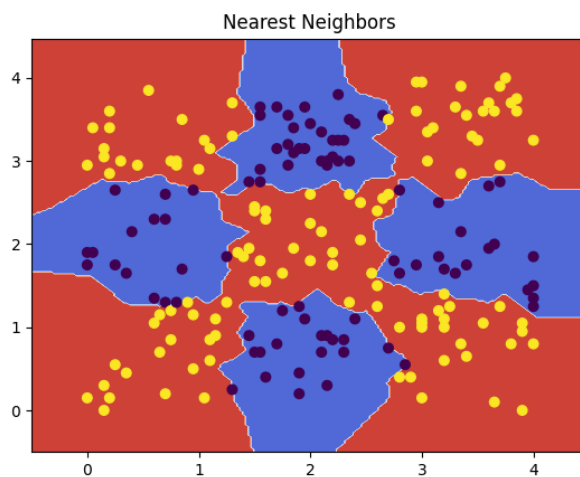
The decision tree learns a set of axis aligned splits that carve the plane into rectangular regions that alternate between the two classes. It captures the rough structure that there is a blue region in the center and red regions around it, but the boundaries are quite blocky and sensitive to the exact split positions which means moderate variance and some overfitting to the training set.



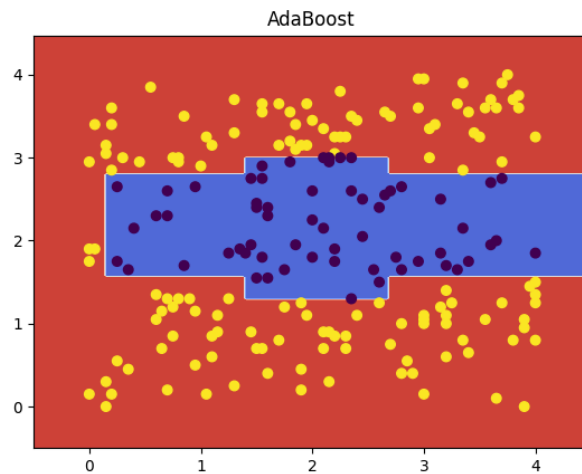
The random forest averages many different trees, so the boundaries are still built from axis aligned cuts but look smoother and less jagged than the single tree. It gets the same basic pattern as the decision tree while reducing the impact of any single noisy split so it should generalize slightly better and usually achieves a higher test accuracy than the single tree.



The logistic regression plot shows a nearly uniform decision region with almost no visible separation between the two classes. This happens because logistic regression uses a linear decision boundary in the original feature space and the classes here are not linearly separable. As a result it underfits badly and will have the lowest accuracy among the methods we tested.



The nearest neighbors classifier produces very irregular decision regions that closely wrap around the training points of each class. It is highly flexible and can fit complicated shapes so it captures the central cluster and the surrounding regions but it is also very sensitive to local noise. This method has low bias but high variance and may perform well on the training data while being less stable on new points.



The AdaBoost classifier combines many shallow trees into a weighted ensemble. In the plot it learns a wide horizontal band for one class and red regions above and below that band. Compared with the single tree AdaBoost focuses on the hard central region while keeping relatively simple boundaries elsewhere. This gives it a balance between flexibility and smoothness and its performance should fall between logistic regression and the best of the tree based methods.